

# Thomas Wang

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## EDUCATION

New York University, College of Arts and Science, New York, NY

9/2022 - 5/2026

*Bachelors of Arts in Computer Science and Mathematics*

- GPA: 3.6 / 4.0
- Awards: Dean's List 2022-2023, 2024-2025
- Relevant Coursework: Data Structures, Fundamentals of Machine Learning, Group Theory, Basic Algorithms, Operating Systems, Numerical Computing

## PROFESSIONAL EXPERIENCES

Barclays, Whippany, NJ

6/2025 - 8/2025

*Technology Analyst*

- Designed and deployed an AI-powered **meeting and video knowledge search chatbot** using **OpenAI Whisper** for speech-to-text, **Qdrant vector database** for semantic search, and **Google Flan-T5** for context-aware summarization, enabling employees to quickly locate answers from recorded **Microsoft Teams meetings** and training videos, projected to save 200-300 hours of support time.
- Developed a **cross-platform project management desktop application** with **Electron**, **Node.js**, and **React**, integrating advanced resource indexing, in-app browser navigation, and fault-tolerant error handling mechanisms; estimated to save approximately 140 hours from run-the-bank employees per week.
- Led the creation of an automated **d3.js mapping of dataflow between critical applications** across **Trading**, **Card Payments**, and **Global Payments**, reducing troubleshooting and error detection time considerably.

Shoptaki, New York, NY

5/2024 - 8/2024

*Full-stack Developer*

- Developed core **SmartID digital passport application** features (login, registration, instant messaging, and medical appointment scheduling pages) using **React** and **React Native**, ensuring seamless user experience.
- Implemented **backend RESTful APIs** for multi-factor authentication, biometrics, messaging, and appointment scheduling with **Python**, **Express.js**, and **MongoDB**; managed approximately 200 new users and processed over 4,000 messages per month.

EMR Technical Solutions, Edmond, OK

5/2023 - 8/2023

*Data Analyst*

- Developed **10-year forecasts of Medicaid enrollment and expenditures** for Oklahoma and the U.S. using **R** and **Python (pandas, statsmodels)**, delivering actionable insights that informed state-level Medicaid funding decisions.
- Built and deployed a **Migaloo blockchain transaction indexer** using **Python**, **Bash**, and **PostgreSQL**; implemented **asynchronous ETL pipelines**, and **Docker containerization**, ensuring high reliability, responsiveness, and scalability.

## PROJECTS

NYC Neighborhood Recommendation Platform

[Link/github](#)

- Built and deployed a full-stack AI application that ranks **71 NYC neighborhoods** by **demographic** and **competitive** fit for a user's prospective business, with a **Next.js** frontend on **Vercel** and a **FastAPI** backend on **Railway**, connected via typed **REST API** with environment-based **CORS** and **health-check monitoring**.
- Engineered unsupervised **K-means clustering** pipeline over **72 features** spanning demographics, foot traffic, business density, safety metrics, and transit access, with interactive K-selection and feature-range controls across **CDTA** geographies, displayed as an **interactive GeoJSON** map.
- Developed a **semantic ranking engine** that scores neighborhoods against a business profile using vector **embeddings**, with **HyDE** and **Supabase**-backed retrieval, as well as a **competitive-saturation ratio**, with an Anthropic Claude agent generating natural-language rationale.

AI-Powered Document Analysis Platform

1/2025 - 6/2025

- Developed an AI-driven document analysis platform with **React**, **Node.js**, and **Express.js**, incorporating **sentiment analysis**, **NME**, and **LangChain**-based **Retrieval-Augmented Generation (RAG)** using **Voyage AI models**, **Google Natural Language API**, as well as **ChromaDB** for intelligent document parsing, contextual response generation, and interactive summarization of PDF, Markdown, and Word documents.
- Built scalable microservices architecture with **Redis caching**, **RabbitMQ message queuing**, and **Firebase** for data persistence, achieving improved efficiency across 10+ API endpoints.
- Deployed production-ready application using modern DevOps practices including **Docker containerization**, **Kubernetes orchestration**, **Jenkins CI/CD pipeline**, and multi-cloud deployment on **Vercel** and **Google Kubernetes Engine** with **NGINX load balancing**.

## SKILLS

**Technical:** Python, Java, AWS, Git, Linux, MongoDB, TypeScript, Node.js, R, PostgreSQL, Docker, Kubernetes, Qdrant, Google Flan-T5, Whisper, Google Cloud NLP, LangChain, RAG, Firebase, Load Balancing, Supabase